

SPECIFICATIONS ($T_J = 25^\circ\text{C}$, unless otherwise noted)						
PARAMETER	SYMBOL	TEST CONDITIONS	MIN.	TYP.	MAX.	UNIT
Static						
Drain-source breakdown voltage	V_{DS}	$V_{GS} = 0\text{V}, I_D = 250\mu\text{A}$	100	-	-	V
V_{DS} temperature coefficient	$\Delta V_{DS}/T_J$	Reference to 25°C , $I_D = 1\text{mA}$	-	0.12	-	$\text{V}/^\circ\text{C}$
Gate-source threshold voltage	$V_{GS(th)}$	$V_{DS} = V_{GS}, I_D = 250\mu\text{A}$	2.0	-	4.0	V
Gate-source leakage	I_{GSS}	$V_{GS} = \pm 20\text{V}$	-	-	± 100	nA
Zero gate voltage drain current	I_{DSS}	$V_{DS} = 100\text{V}, V_{GS} = 0\text{V}$	-	-	25	μA
		$V_{DS} = 80\text{V}, V_{GS} = 0\text{V}, T_J = 150^\circ\text{C}$	-	-	250	
Drain-source on-state resistance	$R_{DS(on)}$	$V_{GS} = 10\text{V}, I_D = 3.4\text{A}^b$	-	-	0.54	Ω
Forward transconductance	g_{fs}	$V_{DS} = 50\text{V}, I_D = 3.4\text{A}^b$	1.3	-	-	S

Fig. 4: Example of a MOSFET's specifications (Source: Vishay IRF510 datasheet)

tem,” says Mrinjal Upadhayay, R&D engineer at Statcon Energia. He continues, “While selecting a MOSFET for any operation we always consider the peak current and voltage and not the operational current.”

The drain-to-source voltage ratings (V_{DS}) should be at least 20% higher than the supply voltage, while the motor load should be higher than the peak load condition. For some cases with high currents, large torque steps, poorly controlled power supplies, etc, a higher margin is required.

The MOSFET should be rated at least for the maximum voltage and the maximum current for the power required. For the motor applications, its current rating should be higher than the peak current. Besides, thermal consideration also plays a vital role in the selection of a MOSFET.

The MOSFETs generate heat due to the drain-source resistance. Thermal constraints tell us about the maximum power that can be dissipated, and the maximum allowable power dissipation drives the selection of the MOSFETs based on the $R_{DS(ON)}$ value.

After determining the necessary voltage rating and $R_{DS(ON)}$, the gate charge (Q_g) value is considered. It determines the speed of the switching and efficiency of the MOSFET. It will switch faster with a device having lower gate drive current than one with a high Q_g .

An example for selection

As an example, let us see how to select a MOSFET for a 12V inverter for a residential application. Let us say the parameters known to us are:

Input - 12V DC (lead-acid battery)

Output - 230V AC

Load - 1000W

Peak load - 2000W

Losses in inverter - 20%

Step 1. Consider the maximum power output at peak load. (While calculating we only consider the maximum output power.)

$P_{\text{max}} = 2000\text{W}$

Total loss is 20% of 2000W, so the total loss is 400W

Step 2. Find out the total load

Total load = Maximum load + Load due to losses

$= 2000\text{W} + 400\text{W}$

$= 2400\text{W}$

Step 3. Calculate peak current and peak load

The input side of the inverter has a lead-acid battery, which will usually have a charge anywhere between 10V and 14V. Therefore, keeping 20% to 40% margin, the input will be approximately 6V to 8V.

Current $= P/V = 2400/6 = 400\text{A}$

Step 4. Calculating the voltage

A minimum of 2x voltage is selected to keep the system stable, because during a short-circuit or peak current a voltage spike is also observed. It is safer to select a higher voltage.

COMPARISON OF Si VS GaN VS SiC MOSFETs

Material	Band gap $E_g(\text{eV})$	Electron saturation velocity $V_s(\text{cm/s})$	Dielectric constant ϵ_r	Critical electric field $E_c(\text{V/cm})$	Thermal conductivity $K(\text{W/cm K})$
Si	1.1	$1 * 10^7$	11.8	$3 * 10^5$	1.5
GaN	3.4	$2.2 * 10^7$	10	$2.2 * 10^6$	1.7
4H-SiC	3.3	$2 * 10^7$	9.7	$2.5 * 10^6$	5

Steps to select a power MOSFET

1. Find out all the parameters of application, such as maximum voltage, maximum current, and operating temperature.
2. Find out the total load of the circuit.
3. Calculate peak current and peak load required for the MOSFET.
4. Find out the efficiency of the system.
5. Now calculate the load with losses.
6. Add a safety factor (depending on the temperature of operation).
7. Check if the device will operate as a bi-directional device or not.